Layout-Preserving Conversion of Mixed Documents

A. K. Mahato and S. Tools — Kathmandu

Abstract. We study how digitally generated and scanned documents can be converted to editable formats while keeping their visual structure. A per-page strategy chooses between the embedded text layer and optical recognition, and a reconstruction step maps positions, fonts and rules into native word-processor constructs.

1. Introduction

Office documents travel between formats many times in their life. Each conversion risks losing structure: columns collapse into one, tables lose their rules, and pictures disappear.

Converters that place every line in a floating text box look right but are hard to edit. Converters that ignore positions are easy to edit but look nothing like the source. We seek a middle ground.

2. Method

Each page is classified by the amount of text in its text layer. Pages with text are analysed from that layer directly; pages without are recognised from their image.

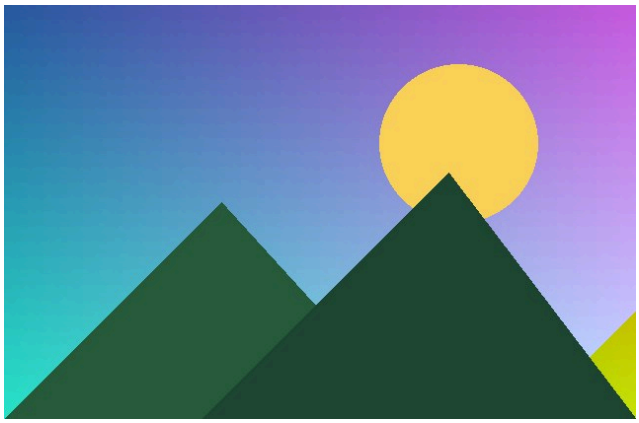


Figure 1. A sample illustration placed inside a column.

Lines are grouped into paragraphs by their spacing and indentation, and columns are found from the gaps between text blocks that run down the page.

3. Results

On a set of business letters, invoices and papers, the reconstructed documents keep their reading order and spacing, and rules and pictures appear where they were.

Remaining differences come from font substitution when the original typeface is not available, which changes line breaks slightly.

4. Conclusion

Combining exact text extraction with careful mapping to native constructs yields documents that are both faithful and editable.